Week 7 Generalised Linear Models

This lab covers linear models and GLMs (logistic and Poisson regression). Work through the sections in order — later questions build on earlier ideas.

For any dataset you're unsure about, `https://rdrr.io/` lists column descriptions. `https://cran.r-project.org/` is also useful, or just Google it.

Some datasets require installing a companion package first (e.g. `install.packages("UsingR")`). The packages used here are:

- `UsingR`: "Using R for Introductory Statistics", Second Edition
- `HSAUR2`: A Handbook of Statistical Analyses Using R (2nd Edition)
- `gamair`: Data for 'GAMs: An Introduction with R'
- `Sleuth3`: Data Sets from Ramsey and Schafer's "Statistical Sleuth (3rd Ed)"
- `ISLR`: Data for an Introduction to Statistical Learning with Applications in R

Some of these books are quite old, but the datasets are still valuable for illustrating the methods.

1 Linear Models

1.1 Drinking Beer

It is well known that the more beer you drink, the more your blood alcohol level rises. Suppose we have the following data on student beer consumption

Student	1	2	3	4	5	6	7	8	9	10
Beers	5	2	9	8	3	7	3	5	3	5
BAL	0.10	0.03	0.19	0.12	0.04	0.095	0.07	0.06	0.02	0.05

Enter the data as two vectors, then make a scatterplot and fit a regression line using `lm()`. The response variable is BAL and the predictor is Beers.

Test the hypothesis $H_0 : \beta_1 = 0.02$ against the one-sided alternative $H_1 : \beta_1 < 0.02$ (i.e. that each additional beer raises BAL by *less* than 0.02). Use the `summary()` output and construct a t-statistic manually, or use `confint()`.

Likelihood Ratio Test

Alternatively, we can frame this as a Likelihood Ratio Test by comparing two nested models:

$$M_0 : \text{BAL}_i = \beta_0 + 0.02 \cdot \text{Beers}_i + \varepsilon_i$$

$$M_1 : \text{BAL}_i = \beta_0 + \beta_1 \cdot \text{Beers}_i + \varepsilon_i$$

where M_0 restricts the slope to 0.02 and M_1 estimates it freely. In `R`:

```
m0 <- lm(BAL ~ 1 + offset(0.02 * Beers))
m1 <- lm(BAL ~ Beers)
anova(m0, m1, test = "LRT")
```

If the test is significant, we reject the hypothesis that $\beta_1 = 0.02$.

Compare the coefficients obtained from `lm()` with those from a direct MLE approach. They should agree.

1.2 Hubble Constant

The package `gamair` has a dataset called hubble `https://rdrr.io/cran/gamair/man/hubble.html`

```
data(hubble, package = "gamair")
```

which takes information about on distances and velocities of 24 galaxies containing Cepheid stars, from the Hubble space telescope key project to measure the Hubble constant.

The Hubble constant is estimated to be defined by the equation (Hubble's law)

$$v = H_0 D$$

where H_0 is Hubble's constant.

Can you fit a simple linear regression model to get an estimate for Hubble's constant? (The University of Chicago puts it at around 69.8 km/s/Mpc, though other measurements have given higher values.) Be careful about the intercept — Hubble's law says $v = H_0 D$ with no constant term, so you should fit a *zero-intercept* model. In R, you can suppress the intercept with `lm(y ~ x - 1)` or equivalently `lm(y ~ 0 + x)`.

Is yours close enough? Do a hypothesis test ($\alpha = 0.05$) to see if 69.8 is a reasonable value based on your model.

Check the confidence interval for the value.

Age of Universe

It is said that the age of the universe can be estimated by $\frac{1}{H_0}$, because of the units in our model we need to do some conversion to see this

```
Mpc <- 3.09 * 10^19
ysec <- 60^2 * 24 * 365.25
Mpcyear <- Mpc / ysec
1 / (coef(model) / Mpcyear)
```

Should give an estimate of about 12.8 billion years, and you thought my lectures were long.

1.2.1 If you've Time - Leverage

An observation is considered to have high leverage if it has a value (or values) for the predictor variables that are much more extreme compared to the rest of the observations in the dataset.

We can get this with hatvalues

```
hatvalues(model)
```

2 of the observations have a higher leverage value, so maybe they're outliers?

Remove the observations for galaxies having leverage greater than 0.08 and refit the zero intercept model. What is the estimated age of the universe from this model?

1.3 Baby Birthweight

The library UsingR has a dataset called babies

```
data(babies, package="UsingR")
```

Explore this dataset. Then do a multiple regression of birthweight (`wt`) with regressors the mother's age (`age`), weight (`wt1`) and height (`ht`).

Is there a simpler form of the model that should be used? Examine the p-values in `summary()` to identify any non-significant predictors, then confirm your decision with a Likelihood Ratio Test. In R, `anova(model1, model2, test="LRT")` (or `lrtest()` from the `lmtest` package) will do this comparison for nested models.

2 Logistic Regression

2.1 Keeping Birds

I mentioned it in class so I have to put it in the lab!
The Sleuth3 package contains this dataset (you may need to install it)

```
library(Sleuth3)
data(case2002)
```

Build a logistic regression model on this dataset using `glm(..., family=binomial)` and summarise the output. Which predictors are statistically significant?

Pay close attention to how R has encoded the categorical variables. By default it treats the first level (alphabetically) as the reference category (coded 0). Check `levels(case2002$LC)` and `levels(case2002$BK)` to confirm which level is which — does 0 mean "has cancer" or "does not have cancer"? Same for the Birds variable: what does `BK = "Bird"` vs `"NoBird"` actually represent in the model?

Compare the coefficients from `glm()` with those obtained by maximising the log-likelihood directly using `optim()`. Write the Bernoulli log-likelihood yourself and check they match.

1. What is the odds-ratio for somebody that keeps birds?
2. What is the odds-ratio for an extra year of smoking?
3. What is the odds-ratio for an extra 10 years of smoking?

Hint: recall that odds ratios come from exponentiating the model coefficients — `exp(coef(model))`. For question 3, think about what multiplying the coefficient by 10 does before exponentiating.

A Note on Odds-Ratio

A common misinterpretation is saying an Odds-Ratio of 4 means that you are 4 times more likely if you do a certain thing. This is not the case, as it is not the underlying probability increasing by a rate of 4.
From Wikipedia

> If we flip an unbiased coin, the probability of getting heads and the probability of getting tails are equal — both are 50%. Imagine we get a biased coin that makes it two times more likely to get heads. But what does "twice as likely" mean in

terms of a probability? It cannot literally mean to double the original probability value, because doubling 50% would yield 100%. Rather, it is the odds that are doubling: from 1:1 odds, to 2:1 odds. The new probabilities would be 66% for heads and 33% for tails.

Also from Wikipedia

> It is standard in the medical literature to calculate the odds ratio and then use the rare-disease assumption (which is usually reasonable) to claim that the relative risk is approximately equal to it. This not only allows for the use of case-control studies, but makes controlling for confounding variables such as weight or age using regression analysis easier and has the desirable properties discussed in other sections of this article of invariance and insensitivity to the type of sampling.

2.1.1 Building Intuition: How do ORs modify absolute risk?

If you have a 1% risk of lung cancer, what does a 8% increase in odds mean? How about a 100% or 200% increase in odds?

```
change_in_prob <- function(orig_prob, OR){
new_odds <- orig_prob / (1-orig_prob) * OR
return( new_odds/(1+new_odds) -orig_prob)
}
change_in_prob(orig_prob=.01, OR=1.08)
```

Try plotting `change_in_prob` over a range of OR values (e.g. 1 to 5) for a few different baseline probabilities. Notice how the *absolute* change in probability depends heavily on where you start.

2.1.2 Best Model - Ratio Test

Is a model with fewer parameters better? Try dropping predictors with high p-values in the summary and apply a likelihood ratio test to compare nested models (`anova(model_full, model_reduced, test="LRT")`). Also compute and compare AIC and BIC with `AIC()` and `BIC()` — do all three criteria agree on the preferred model?

2.2 Predicting Diabetes using the Pima Indian Diabetes Dataset

Dataset: `"Pima.tr"` from the `MASS` package.

1. Load the dataset using `data("Pima.tr", package = "MASS")`. `https://rdrr.io/cran/MASS/man/Pima.tr.html` if you want to see what the columns mean.
2. Build a logistic regression model predicting `type` (diabetes status) using `npreg` (number of pregnancies).
3. Extend the model by including `glu` (glucose concentration) as an additional predictor.
4. Compare the models using a likelihood ratio test.
5. Interpret the coefficients and their significance. Compute and interpret the odds ratios for both predictors.

6. Based on the odds ratios, discuss the effect of `npreg` and `glu` on the likelihood of having diabetes.

2.3 Predicting Default using Credit Data

Dataset: `"Default"` from the `ISLR` package.

1. Load the dataset using `data("Default", package = "ISLR")`. (Personally I think that's a stupid name for a dataset!)
2. Build a logistic regression model predicting `default` using `balance` (credit card balance).
3. Extend the model by including `income` as an additional predictor.
4. Compare the models using Akaike Information Criterion (AIC). Which model fits better?
5. Compute and interpret the odds ratios for `balance` and `income`. What does a unit increase in each variable imply about the probability of default?
6. Do you get similar parameters using MLE?

3 Poisson Regression - Claims Example

Some things to compare with the lecture

```
library(MASS)
data(Insurance)
head(Insurance)
poisson_model <- glm(Claims ~ District + Group + Age + offset(log(Holders)),
                     family = poisson, data = Insurance)
```

Was how I did the poisson_model for the Claims example

3.1 Categorical Variables Differences?

I mentioned about Age.L and Group.L and how R uses orthogonal polynomials for numeric factors, and said it was possible to make it standard dummy variables for this categorical data.

Does this make any difference to the model performance? Try it.

```
Insurance$Age <- factor(Insurance$Age)
contrasts(Insurance$Age) <- contr.treatment(levels(Insurance$Age))
```

forces R to do them as dummy variables when you make a GLM. So rebuild the model after doing this with Age and Group.

Is there any difference between AIC scores?

Why Orthogonal Polynomials?

You will see no difference in scores, so what's the point in the different forms? Remember I am not just interested in making predictions (so for ML it would make no difference which to go with), I am interested in interpreting the coefficients.

- Using the Orthogonal Polynomials showed us the linear relationship as age and engine size increased

- Using the Category Dummy Data idea shows us how the different categories compare to the baseline but we cannot directly say as age increases, number of claims decreases.
- Or that as engine size increases, number of claims increases.

3.2 Offset?

It should be straightforward to build a model without the offset term — just remove `offset(log(Holders))` from the formula.

Why does the offset matter? Without it, the model predicts total claim counts without accounting for the number of policyholders in each group. Including `offset(log(Holders))` *effectively models the* rate *of claims per holder, which is what we really care about.*

Compare the two models on:

- AIC
- BIC
- Residual Deviance

Which model should we go with?